

Status of High-End Performance Packaging (2.5D & 3D) and Co- packaged optics

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Technology & Market Analyst,
Semiconductor Equipment



PRESENTATION OUTLINE

Status of High-End Performance Packaging (2.5D & 3D) & CPO

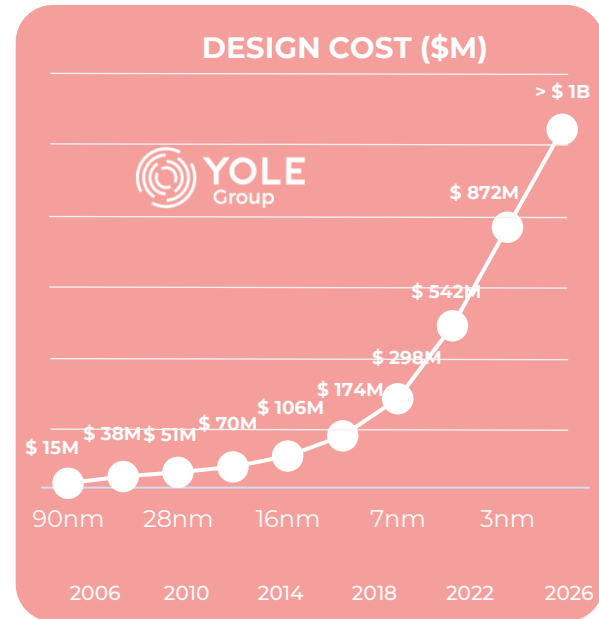
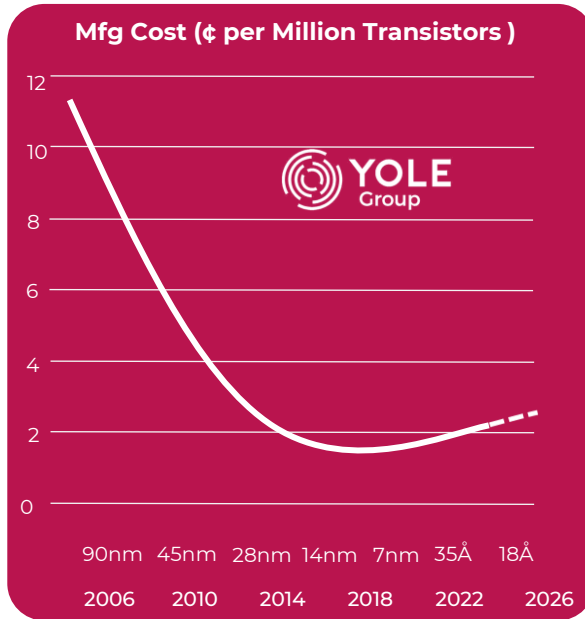
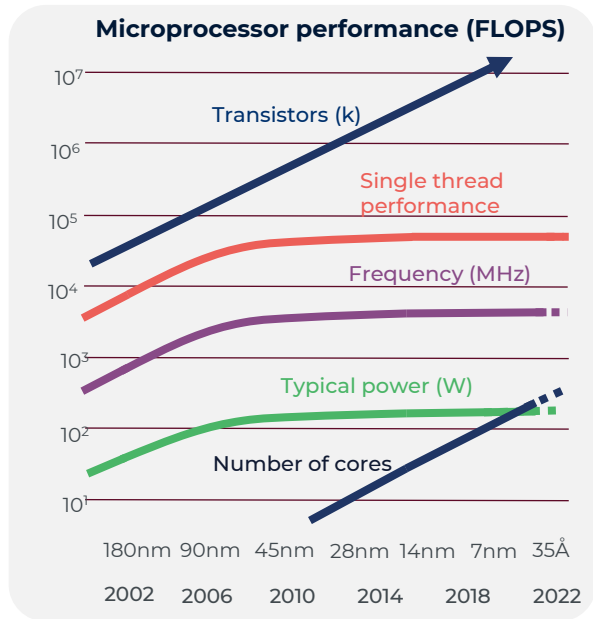
- Global Trends of Semiconductor Industry & Advanced Packaging
- High-End Performance Packaging
- Co-Packaged Optics
- Q&A

GLOBAL TRENDS OF SEMICONDUCTOR INDUSTRY & ADVANCED PACKAGING



LEADING-EDGE NODES – STEEP INVESTMENT COST

Semiconductor FE cost is increasingly hefty: need alternative in advanced packaging



- Moore's Law still holds true but is decelerating due to the atomic limit while at the same time, rising design costs and manufacturing complexity have led to a longer time-to-market.
- The cost of building new fabs for the most advanced nodes is exorbitant and often not worth the investment for many applications.
- Hence the industry's approach lies in the adoption of the chiplets and utilizing different AP technologies to design and manufacture the latest SoCs using SiPs, which can reduce costs and time-to-market while enhancing system performance.



01 Partitioned die
➤ more dies per wafer

02 Higher yield
➤ Optimized cost

03 Finer bump or pad pitch
➤ Higher density

04 Optimization node per chiplet

05 Faster time-to-market



How to realize chiplet & heterogeneous integration?



An Advanced Packaging platform such as 2.5D/3D, fan-out or high-density flip-chip is needed.

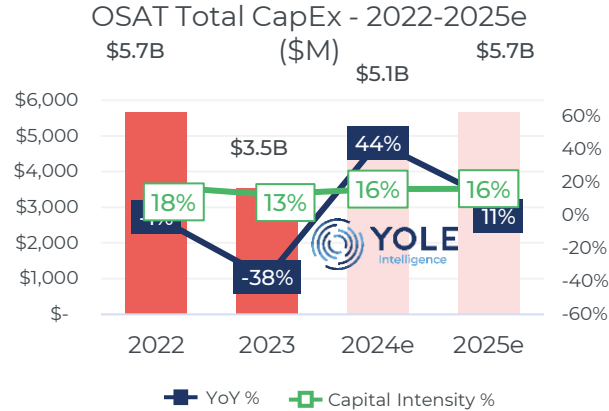
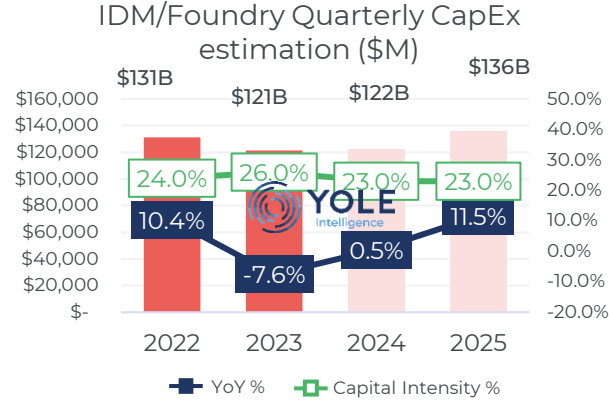
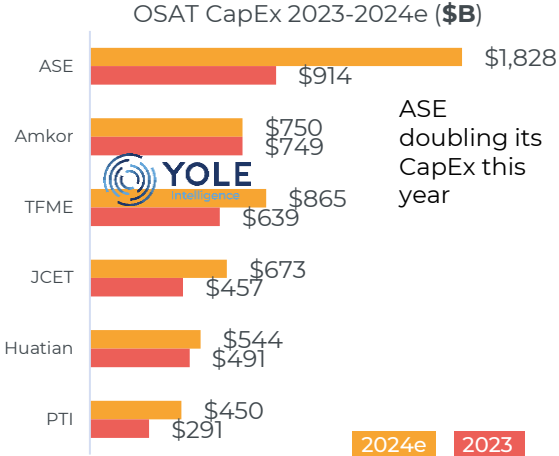
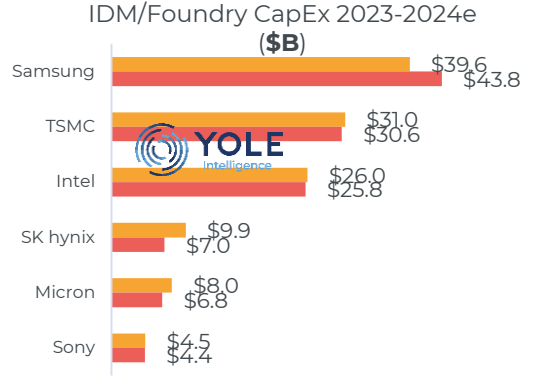


FC on organic substrate
FC on organic with RDL (UHD FO)
Mold Interposer + Si Bridges
Si Interposer
Si bridges
D2W/ W2W hybrid bonding (3D SoC)

CAPEX HIGHLIGHTS FOR TOP PACKAGING PLAYERS



Note: ASE CapEx includes SPIL and excludes USI
All data is based on companies' official results, Q2-24 data and onwards is estimated.



HIGH-END PERFORMANCE PACKAGING

HIGH-END PERFORMANCE PACKAGING MARKET TRENDS

Value creation opportunity for high computational applications



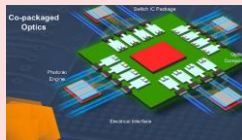
Data center networking



Server – Cloud Computing



Switches – HPC



Co-packaged Optics

High-Performance Computing



Workstations

AI-powered Content



High-Performing
Desktop/Laptop



Big-data AR/VR

Autonomous Vehicles



AI-enabled Passenger Cars



Robotaxi



Autonomous Driving

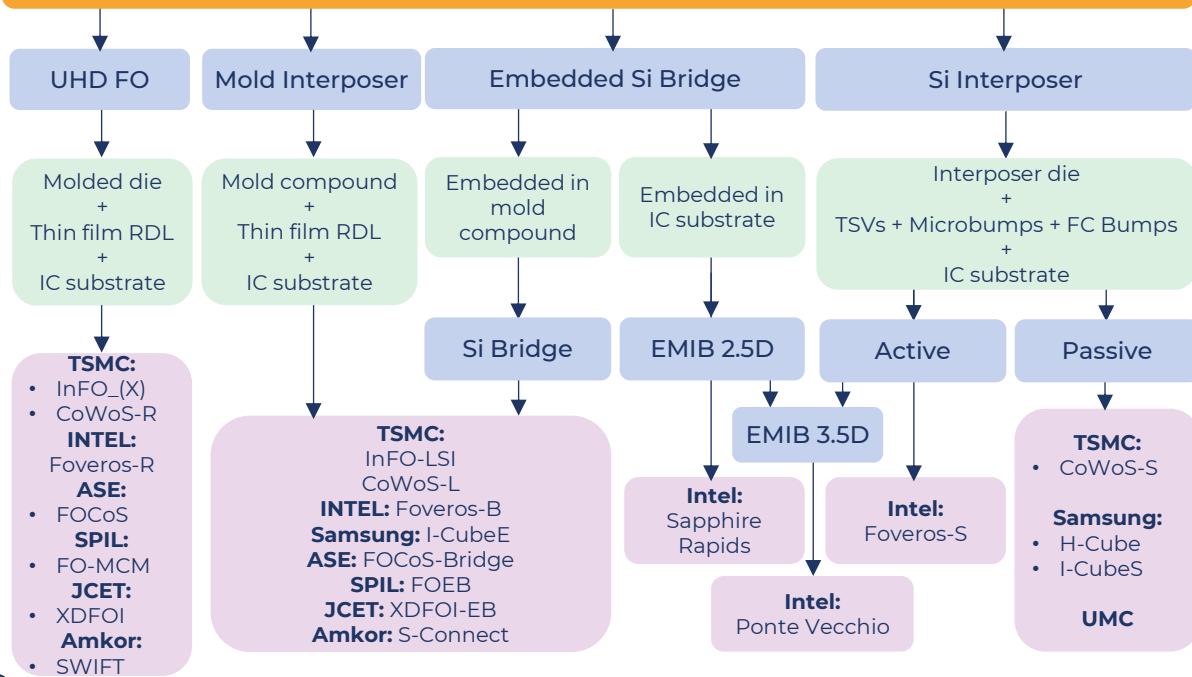
HIGH-END PERFORMANCE PACKAGING: ALL PLATFORMS

Yole's classification for 2.5D and 3D

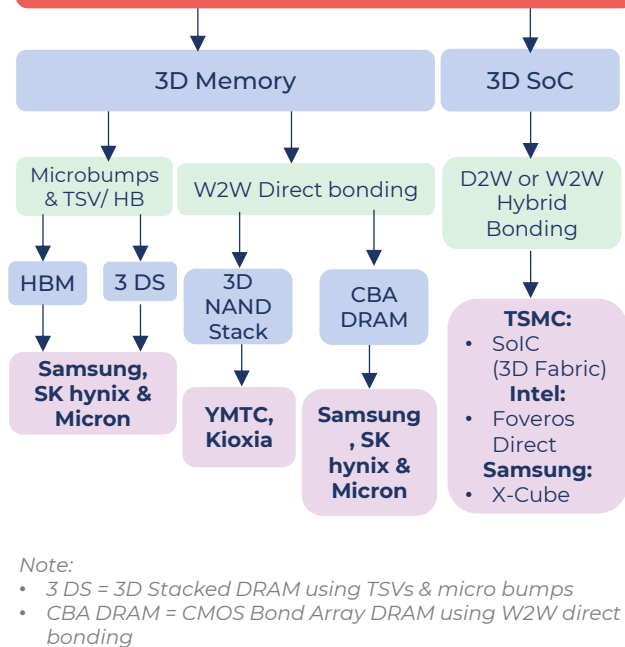


High-End Performance Packaging

2.5D



3D



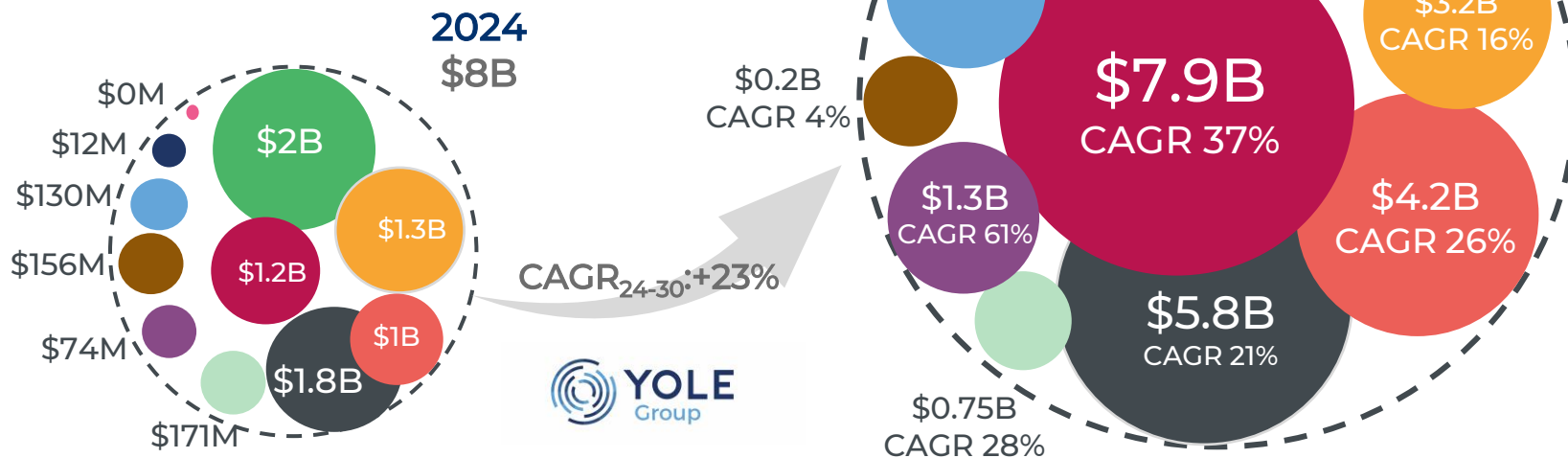
Note:

- 3 DS = 3D Stacked DRAM using TSVs & micro bumps
- CBA DRAM = CMOS Bond Array DRAM using W2W direct bonding

2024 - 2030 HIGH-END PERFORMANCE PACKAGING MARKET SIZE

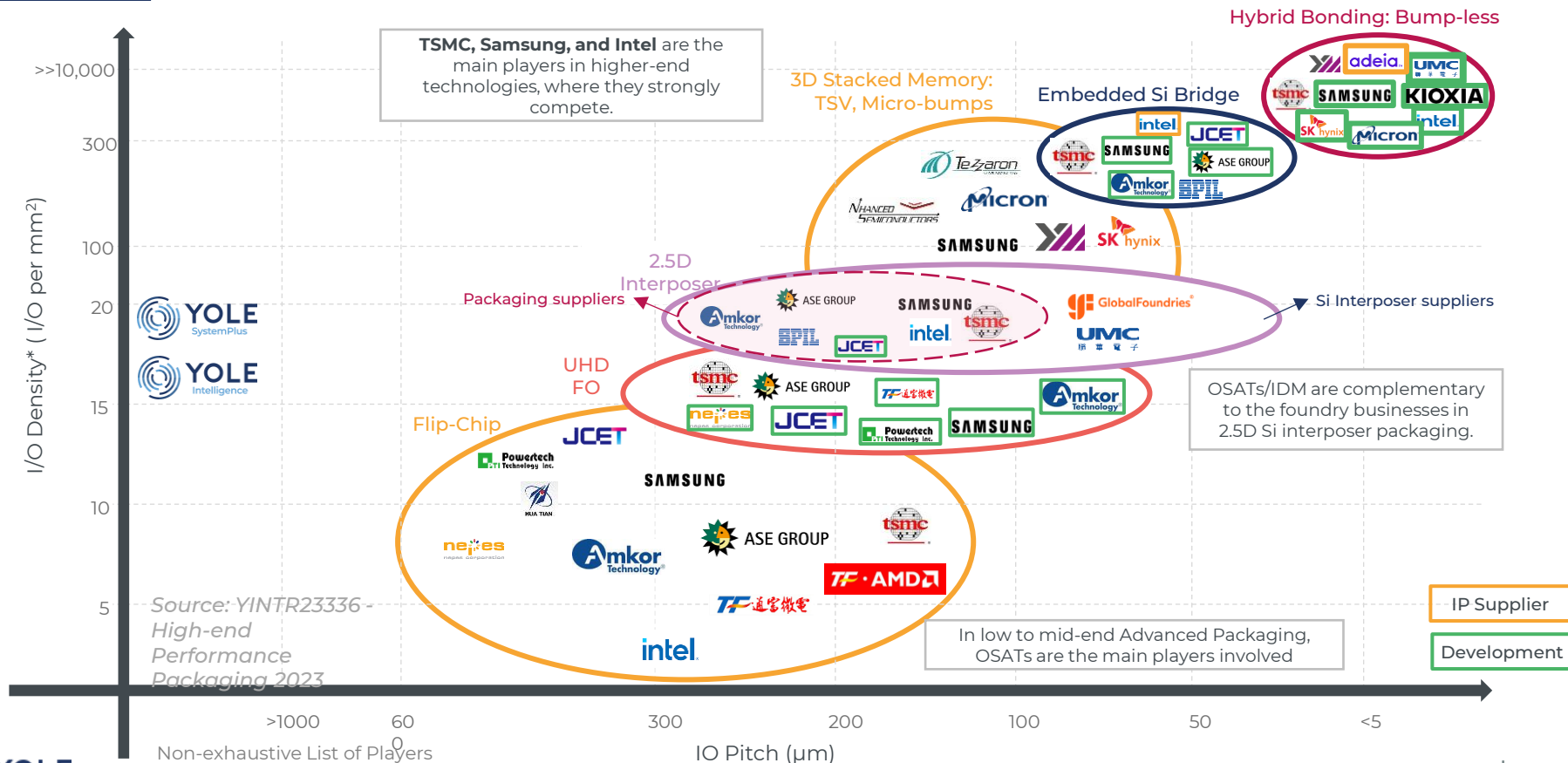
Split by technology

- HBM
- 3DS
- Si Interposer
- Mold Interposer
- 3D SoC
- 3DNAND
- Active Si Interposer
- UHD FO
- CBA DRAM
- Co-EMIB
- Si Bridge



HIGH-END PACKAGING – SUPPLY CHAIN

Mapping of players based on technology

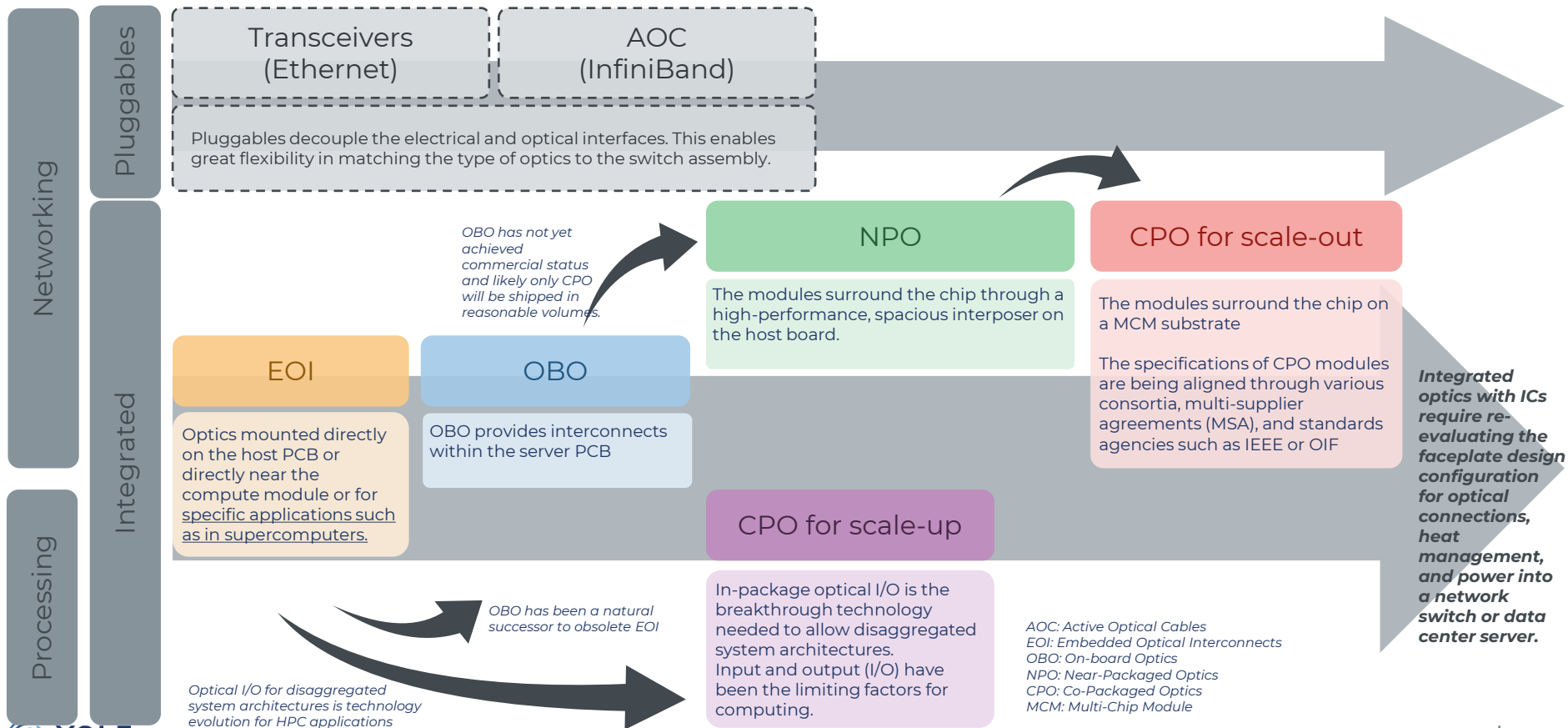


CO-PACKAGED OPTICS



DATAKOM APPLICATION TRENDS

Evolution of integrated optics paralleling pluggable optics

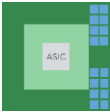


CPO MARKET

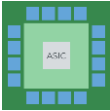
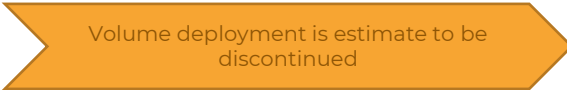
Technologies – Time to market



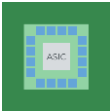
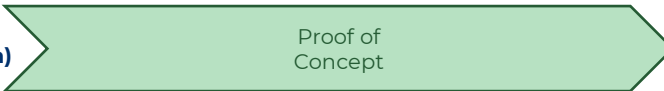
Scale-out



**EOI
(Supercomputers)**



**NPO
(Networking: switch)**

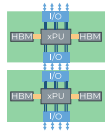


**CPO for Networking
(Switch)**

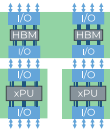
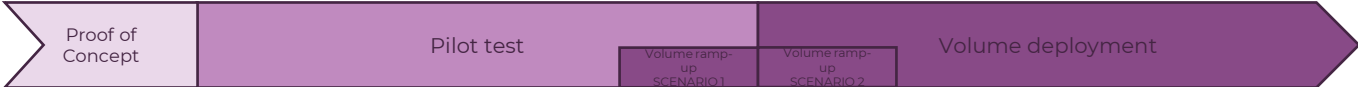


EOI: Embedded Optical Interconnects
OBO: On-board Optics
NPO: Near-Packaged Optics
CPO: Co-Packaged Optics

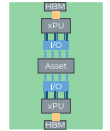
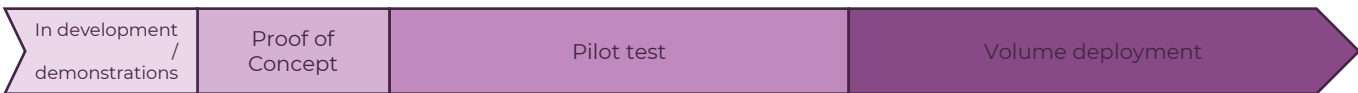
Scale-up



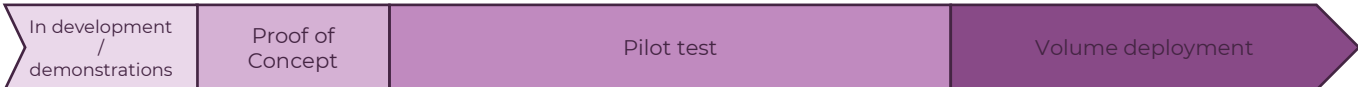
**CPO for
HPC: AI/ML**



**CPO for
Disaggregated
Memory)**



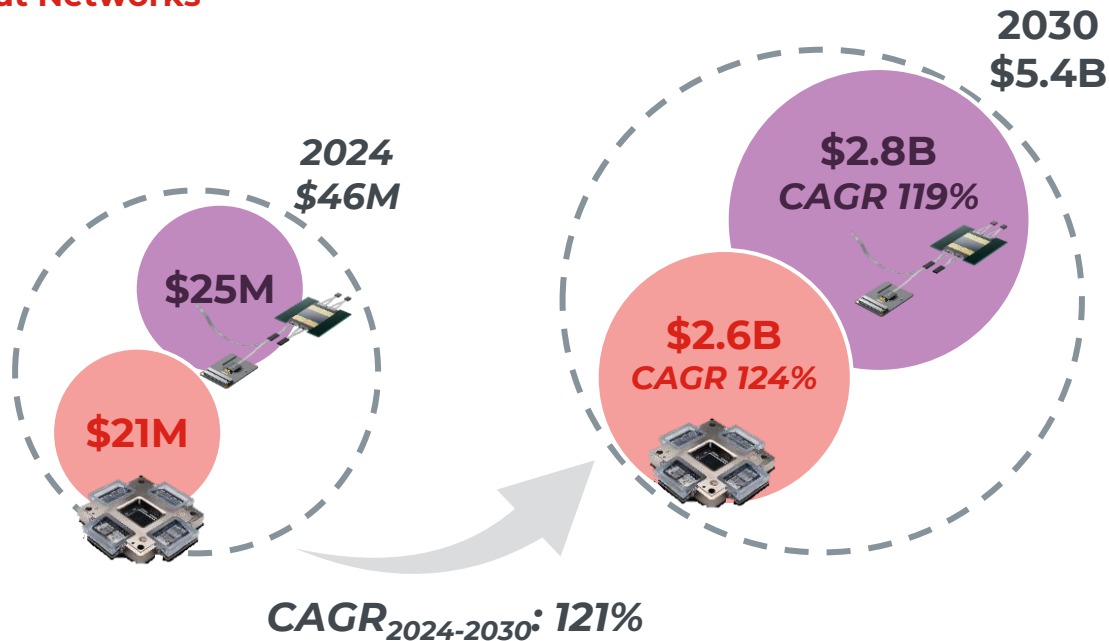
**CPO for
Disaggregation:
Resource pooling**



CPO MARKET

Datacom CPO revenue growth forecast: by application (2024 vs. 2030)

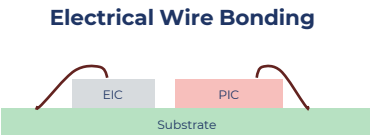
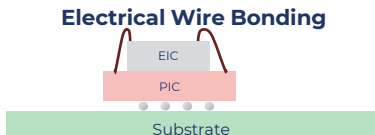
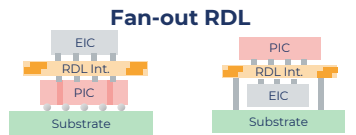
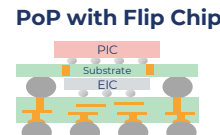
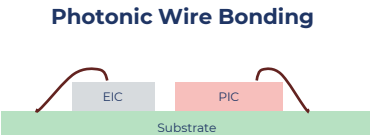
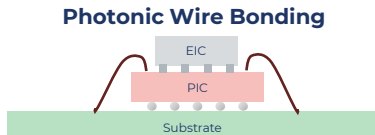
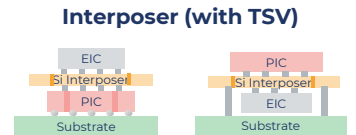
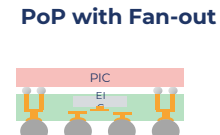
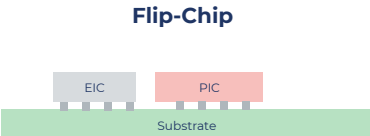
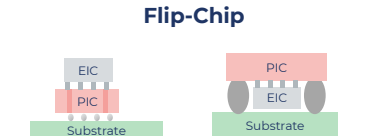
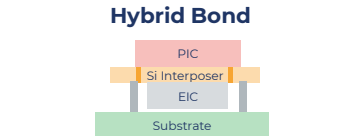
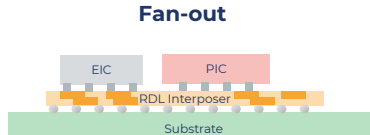
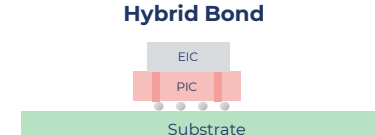
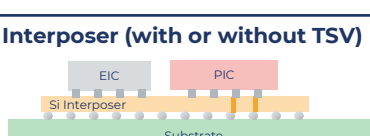
- CPO for Scale-Up Networks
- CPO for Scale-Out Networks



CPO - OPTICAL ENGINE

Heterogeneous Integration of EIC and PIC



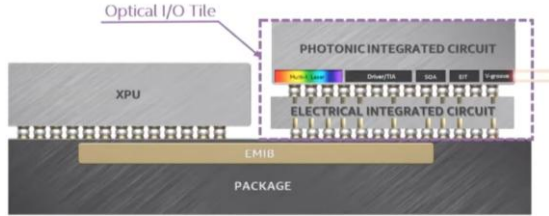
2.0D or 2.5D	2.5D and 3.0D		
EIC and PIC Side by Side	EIC /PIC Stacking	EIC /Interposer /PIC Stacking	Package on Package (PoP)
Electrical Wire Bonding 	Electrical Wire Bonding 	Fan-out RDL 	PoP with Flip Chip 
Photonic Wire Bonding 	Photonic Wire Bonding 	Interposer (with TSV) 	PoP with Fan-out 
Flip-Chip 	Flip-Chip 	Hybrid Bond 	Which HI solution will win? • Lower cost • Higher heat dissipation • Reliable interconnect solution Share with us the packaging solution developing in your company.
Fan-out 	Hybrid Bond 	PIC as interposer? Embedding in substrate? Use Si bridge? Glass substrate/ interposer? Multiple EIC/PIC? 3D monolithic?	
Interposer (with or without TSV) 	Multiple advanced packaging technologies are proposed to integrate PIC and EIC. Which type of heterogeneous Integration solution will win with more adoption? Yole's semiconductor packaging team is keeping an eye and updating the dynamics in the industry.		

PHOTONIC PACKAGING

Technology offered by companies



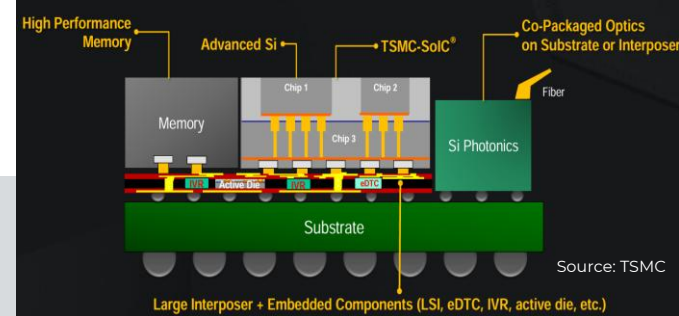
Intel



Source: Intel

Intel CPO connects to OE to XPU via EMIB packaging technology while also connecting the EIC and PIC in a 3D stack with TSVs.

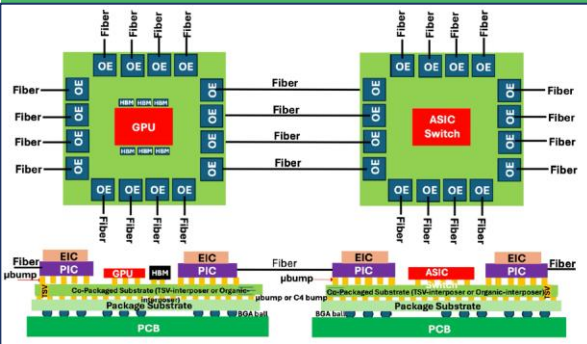
TSMC's COUPE integrates EIC + PIC + Fiber Attach Unit (FAU) + Advanced Coupler (AC) using 3D Fabric solution (CoWoS + SoIC).



Source: TSMC

TSMC

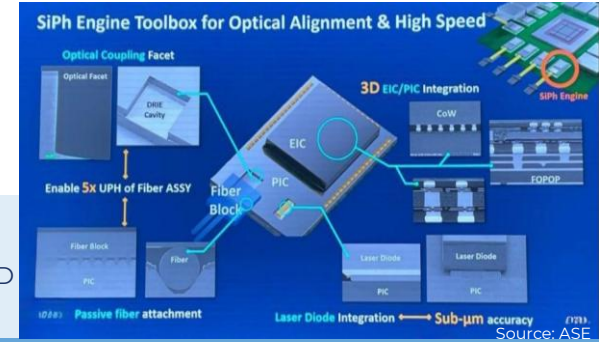
Nvidia



Source: Nvidia

Nvidia is planning to integrating OE in its high-performance GPU products, starting in 2025.

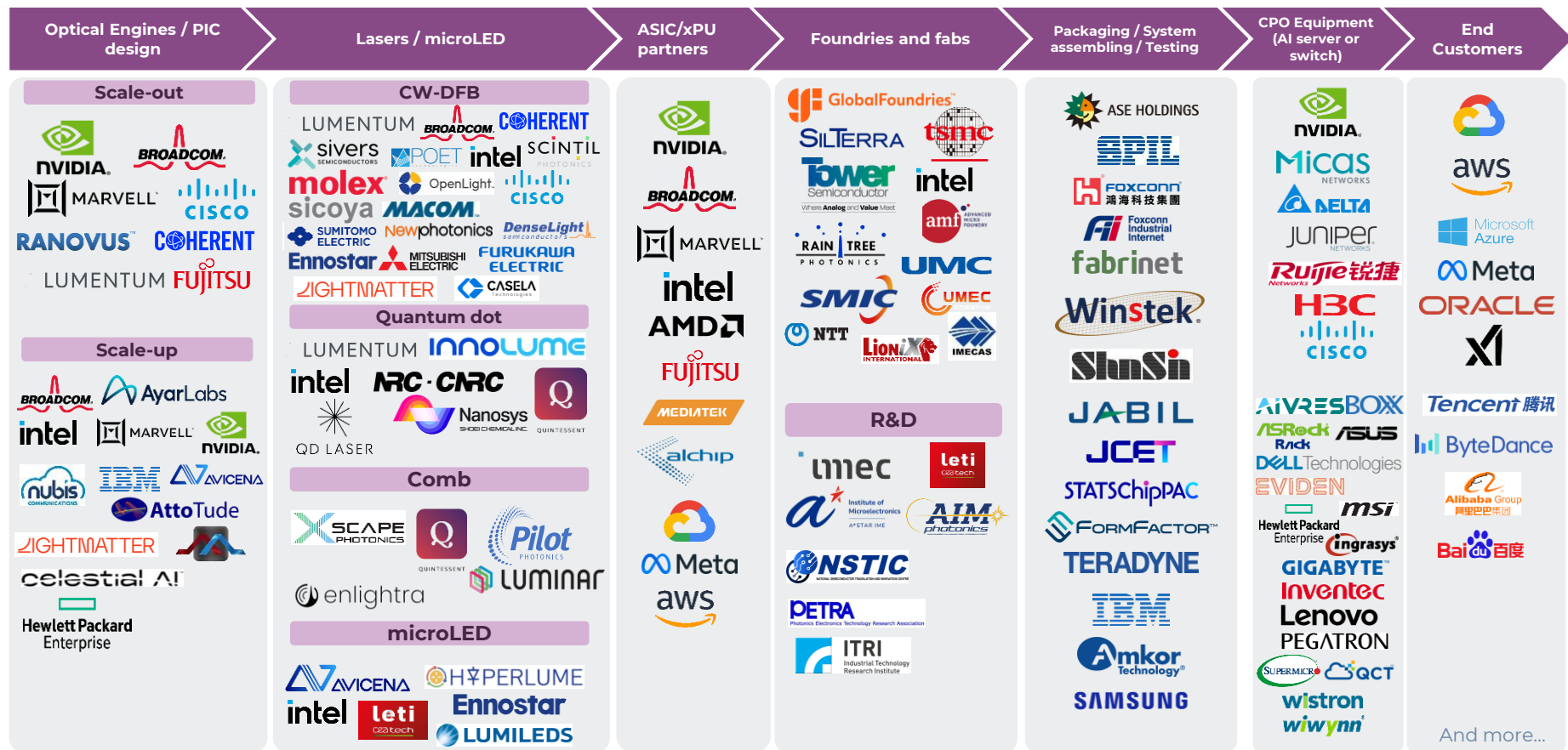
ASE is using VIPack™ advanced packaging platform for CPO. Adopting high accuracy laser die bonding, 2.5D/3D with TSV/FO/CoW packaging solutions.



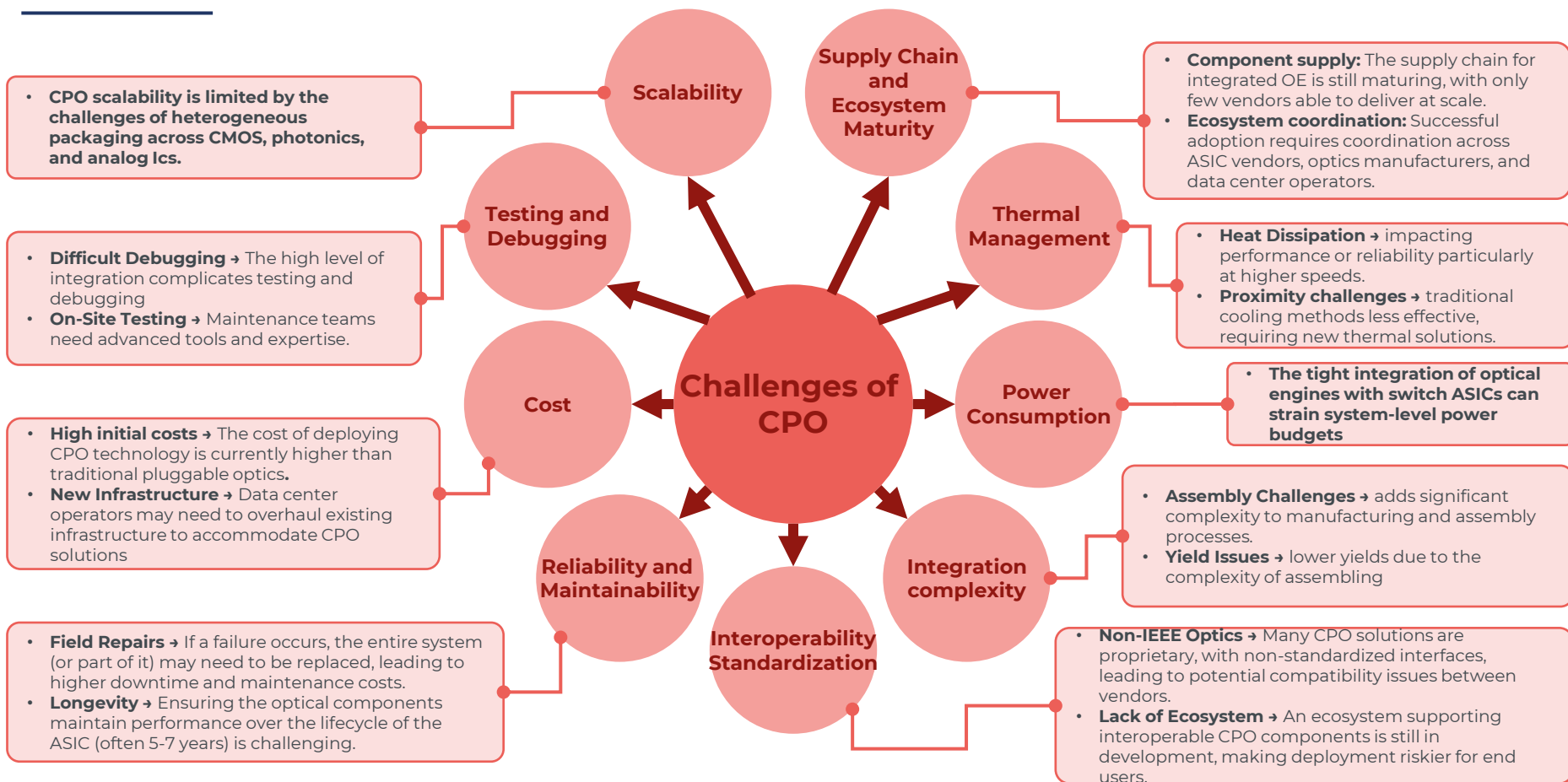
Source: ASE

ASE

SILICON PHOTONICS VALUE CHAIN FOR CO-PACKAGED OPTICS



*non-exhaustive list





THANK YOU

Q&A

INTELLIGENCE TO SHAPE YOUR TOMORROW

Yole Group

Corporate Presentation





REPORTS

Insight

- › Yearly reports
- › Market, technology and strategy analysis
- › Supply chain changes analysis
- › Reverse costing and reverse engineering

Format

- › PDF files with analyses
- › Excel files with graphics and data

Topics

- › Photonics, Imaging & Sensing
- › Lighting & Displays
- › Power Electronics & Battery
- › Compound Semiconductors
- › Semiconductor Manufacturing and Packaging

115+ reports per year

MONITORS

Insight

- › Quarterly updated market data and technology trends in units, value and wafer
- › Direct access to the analyst

Format

- › Excel files with data
- › PDF files with analyses graphs and key facts
- › Web access (to be available soon)

Topics

- › Advanced Packaging
- › Application Processor
- › DRAM & NAND
- › Compound Semiconductor
- › CMOS Image Sensors
- › Micro-controller
- › Semiconductor Test Equipment

7 different monitors quarterly updated

TRACKS

Insight

- › Teardowns of phones, smart home, wearables and automotive modules and systems
- › Bill-of-Materials
- › Block diagrams

Format

- › Web access
- › PDF and Excel files
- › High-resolution photos

Topics

- › Consumer: Smartphones, smart home, wearables
- › Automotive: Infotainment, ADAS, Telematics

**205+ teardowns per year
Daily updates**

CUSTOM SERVICES

Insight

- › Specific and dedicated projects
- › Strategic, financial, technical, supply chain, market and other semiconductor-related fields
- › Reverse costing and reverse engineering

Format

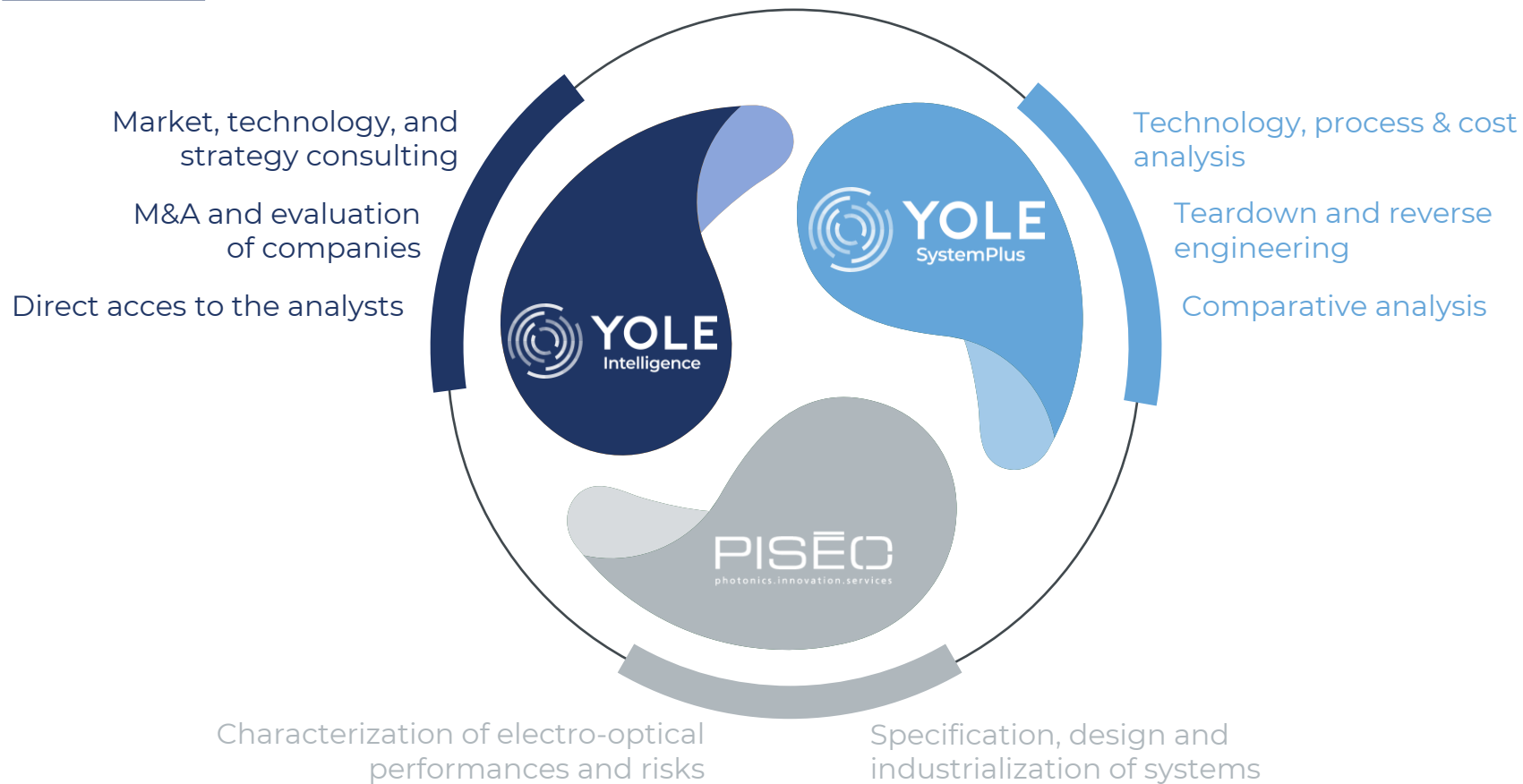
- › PDF files with analyses
- › Excel files with graphics and data

Topics

- › Photonics, Imaging & Sensing
- › Lighting & Displays
- › Power Electronics & Battery
- › Compound Semiconductors
- › Semiconductor Manufacturing and Packaging

190 custom projects per year

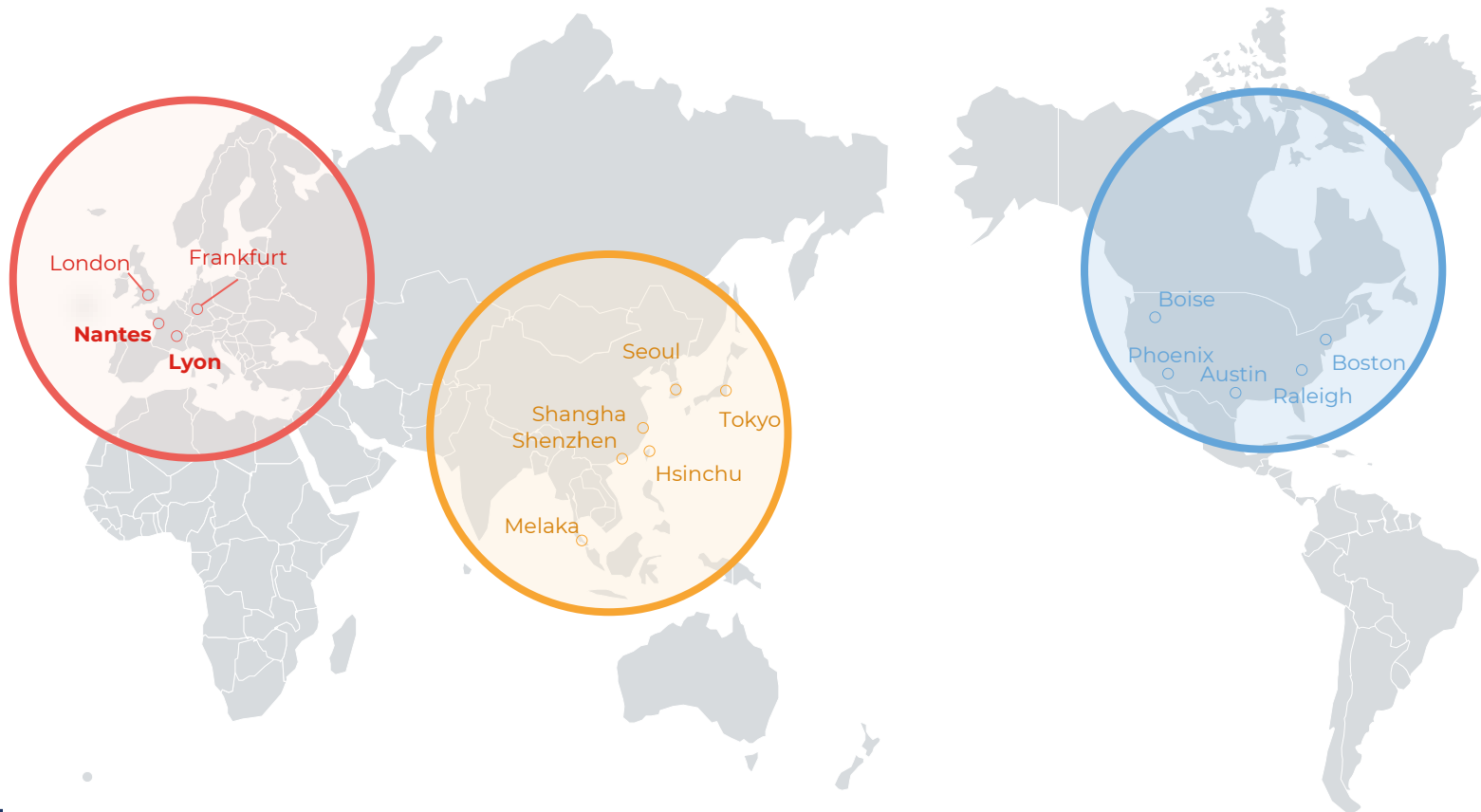
YOLE GROUP'S MAJOR ACTIVITIES PER ENTITY



A WORLDWIDE PRESENCE



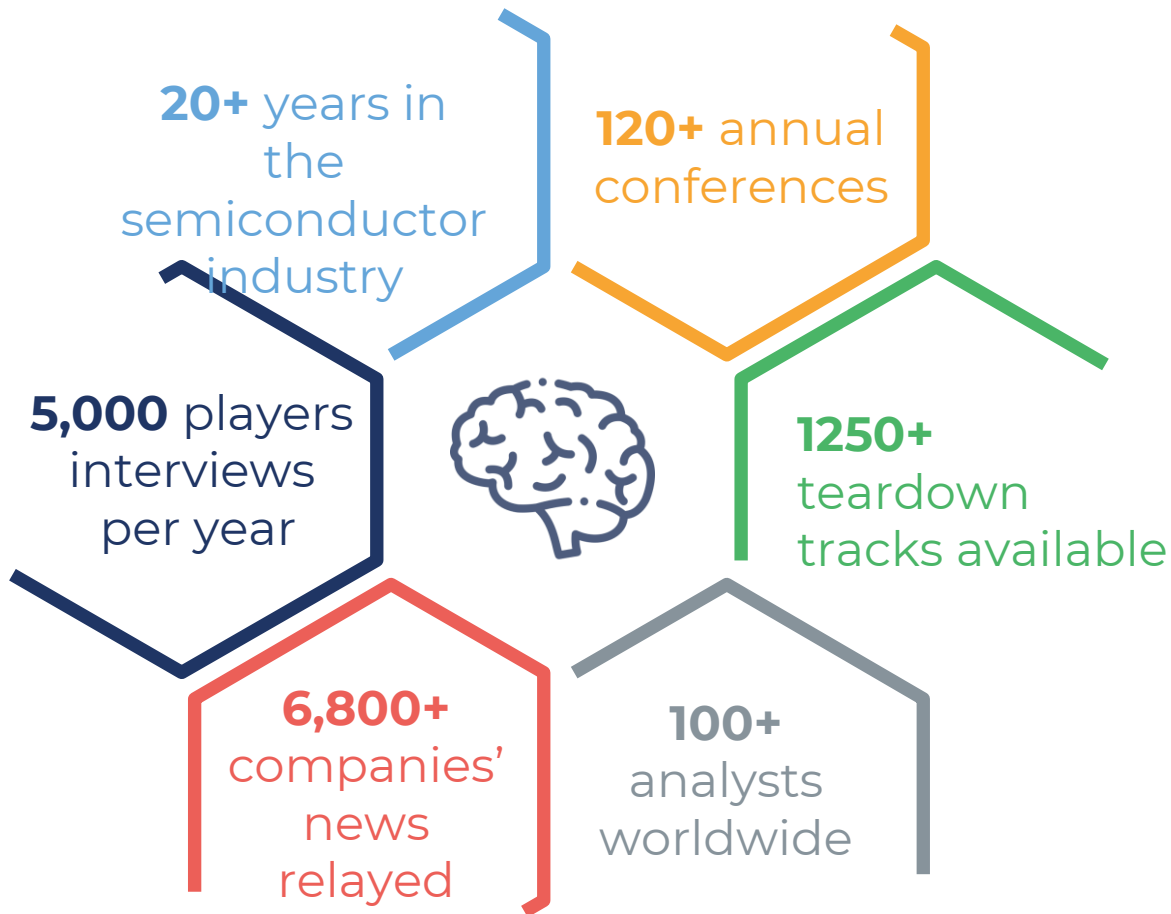
180+ collaborators in 9 different countries



A WIDE RANGE OF INFORMATION SOURCES



Our unique position allows us to obtain detailed and accurate information to meet your needs.





MARKETING EXCELLENCE AND BEST-IN-CLASS NETWORK

- **Market segmentation**
 - Per application
 - Per technical needs
 - Per technology adoption and supply chain's tendencies
- **Primary research and direct interviews with key players**



"Thanks to its unique **semiconductor market intimacy**, its understanding of the **industrial environment** and its vision on **future technologies** adoption, Yole Group supports its customers at every stage of their growth"



BOTTOM-UP, TOP-DOWN AND INDUSTRIAL EXPERTISE

- **Top-down**
 - End market demand analysis
 - Market forecasts at system and component levels down to wafer and equipment
- **Bottom-up**
 - Ecosystem analysis
 - Consolidate industrial players' revenue at component, module and system levels
- **Industrial experts in all our fields of investigation**

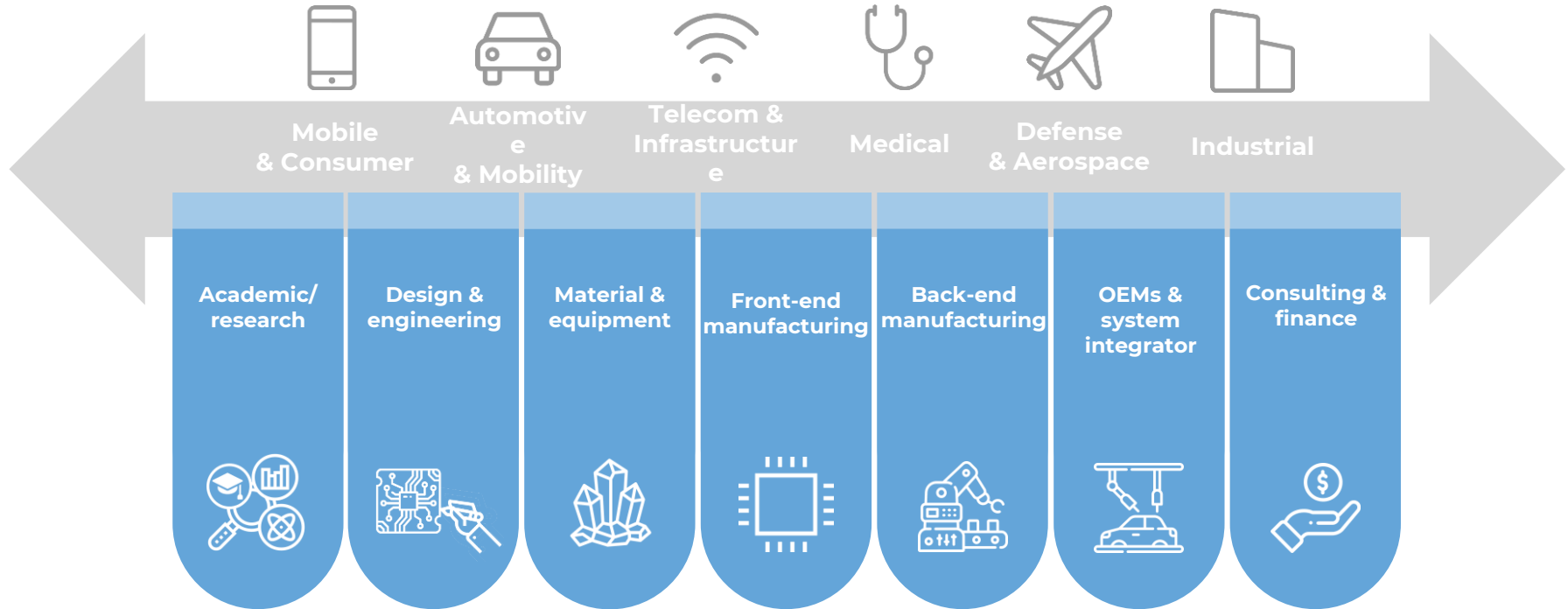
STATE-OF-THE-ART TECHNOLOGY AWARENESS

- **Technology analysis**
 - Competitive landscape and technology comparison
 - Reverse costing
 - Reverse engineering
- **Technology life cycle**
 - Development cycles
 - Supply chain adoption
 - HV manufacturing and evolutions
- **Performance testing and analysis**

OUR NETWORK IS THE ENTIRE SUPPLY CHAIN ACROSS 6 MARKETS



6 KEY MARKETS





REPORTS, MONITORS & TRACKS

NORTH AMERICA

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